

Calibrated Intent Compilation and Faithful Explanation for Human-Supervised Agricultural Robotic Systems

PhD Position L — GreenFieldData MSCA Joint Doctorate

Partha Pratim Saha

`technical.parth@gmail.com`

Supervisors:

Dr. Genoveva Vargas-Solar (CNRS/UCBLyon1, Lyon)

Prof. Roberto Oberti (Università degli Studi di Milano)

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The Real Problem Is Not “Build a Chatbot for Farmers”

A Field Operator Says...

“Spray more pesticide on the struggling section near the east boundary.”

This single utterance is simultaneously ambiguous over:

- **Spatial:** which section? (NDVI map required)
- **Quantitative:** “more” relative to what baseline?
- **Chemical:** which pesticide, which dosage?
- **Temporal:** safe *now*? (current wind speed?)

Consequences of Getting It Wrong

- Wrong zone → crop destruction
- Wrong chemical → resistance development
- Wrong conditions → environmental harm + regulatory violation

And of Over-Correction

A system that asks for confirmation on every query
→ **alert fatigue** → **system abandonment**

Research challenge: resolve ambiguity with statistically grounded confidence, real-time field context, and explanations operators can audit and trust.

Three Interlocking Scientific Challenges

#1 Calibration Gap

LLM softmax is overconfident on OOD agricultural inputs (Guo et al., ICML 2017).

Need: statistically valid uncertainty, not heuristic thresholds.

#2 Streaming Grounding

Conditions change faster than queries arrive:
wind 2 m/s at 08:00 → 7 m/s at 14:00.

Need: atomically consistent context at query receipt time.

#3 Faithfulness Gap

LLM explanations can sound correct but not trace to actual computation (Lipton 2018).

Need: explanations verifiable by agronomists & EU AI Act audit.

Challenge	State of the Art	Gap	My Contribution
Calibration	Softmax confidence, temperature scaling	Not distribution-free	C1: Conformal CIC
Streaming	Static KB retrieval	No IoT-stream grounding	C2: TSCA
Faithfulness	Post-hoc LLM explanation	Plausible $\neq$ faithful	C3: FAES

Research Questions & Novel Contributions (C1–C4)

Research Questions

- RQ1** Can conformal prediction sets enable principled HITL policies that minimise *both* unsafe autonomy *and* alert fatigue?
- RQ2** Can streaming context aggregation ground NL commands against atomically consistent, time-indexed IoT field state?
- RQ3** Can explanation generation be *measurably faithful* to the actual computational path?
- RQ4** Can Metaflow reproducibility enable comparative evaluation at agricultural field scale?

Novel Contributions

- C1 Calibrated Intent Compiler (CIC)**
Conformal prediction → HITL policy mapping
ACL 2027 / EMNLP 2027
- C2 Temporal Streaming Aggregator (TSCA)**
Kafka + Spark sliding-window context register
VLDB 2027 / AAAI 2028
- C3 Faithful Explanation System (FAES)**
Rule-auditable + KB-traceable + IG attribution
FAccT 2028
- C4 Reproducible AI Infrastructure (RAAI)**
Metaflow DAG for reproducible agri-AI experiments
MLSys 2027

AgriTalk System Architecture: Five Information Boundaries

B#	Boundary	What Happens	Contrib.
B1	Physical → Digital	IoT sensors → raw streams; data quality validation	C4
B2	Streaming → Context	Kafka + Spark 15-min sliding window → field-state register	C2
B3	Language → Intent	NL query → 5-class intent + slots; conformal confidence	C1
B4	Intent → Command	Slot grounding vs EPPO/PPDB/AGROVOC KB; unverified blocked	C3
B5	Command → Robot	HITL gate → ROS2 publication; ABORT path <800 ms edge	C1, C3

Edge Deployment (B5)

Mistral-7B 4-bit GGUF on NVIDIA Jetson Orin
→ ABORT latency target: <800 ms
Same corpus + calibration set as cloud model

Cloud Inference (B3, B4)

Llama-3-8B-Instruct on GPU endpoint
Complex / latency-tolerant queries
Full Integrated Gradients attribution available

Each boundary = a testable failure mode = a research artifact

The Calibration Insight: Conformal Prediction Sets $\rightarrow$ HITL Policy

Why not softmax confidence?

- Overconfident on OOD agricultural commands (Guo et al. 2017)
- No distribution-free guarantees
- Threshold-tuning is manual and brittle

Conformal prediction gives:

$$\mathbb{P}(y \in \mathcal{C}(x)) \geq 1 - \alpha$$

Distribution-free. One calibration forward pass.
Deployable on Jetson Orin.

Prediction Set	Meaning	HITL Decision
{COMMAND}	High confidence	Proceed (safe check only)
{COMMAND, ABORT}	Ambiguous	HITL mandatory
All 5 classes	OOD input	Reject + rephrase

Research Novelty

HITL rate = tunable parameter (via α). We empirically study the **alert-fatigue vs. safety tradeoff** across $\alpha \in [0.01, 0.20]$.

Safety-Asymmetric Extension (Mondrian Conformal)

ABORT class gets stricter coverage ($\alpha_{\text{abort}} = 0.01$) than routine queries ($\alpha_{\text{query}} = 0.10$) — per-class Mondrian conformal prediction.

Human-in-the-Loop Gate: Rule-Auditable, Not Learned

5 Rule-Auditable Trigger Conditions:

- 1 Action $\in \{\text{SPRAY}, \text{ABORT_MISSION}, \text{DOSAGE_CHANGE}, \text{ZONE_OVERRIDE}\}$
- 2 Conformal set non-singleton (≥ 2 intents)
- 3 Any entity unverified against KB
- 4 User role = OBSERVER (read-only)
- 5 Upstream data quality check failed

Each trigger $\rightarrow$ **explicit human-readable audit string**
(no LLM involved in trigger explanation)

Operator Response Options

- YES $\rightarrow$ execute command as proposed
- NO $\rightarrow$ abort; log refusal reason
- MODIFY $\rightarrow$ operator specifies override; system re-grounds

Why Rule-Auditable (Not Learned)?

A learned gate cannot be audited by an agronomist or regulator.

Rule-auditable gates are: version-controlled, explainable by design, testable with unit tests.

HITL is not a limitation — it is the research subject.

RQ1 asks: what is the optimal α for each action class to minimise unnecessary HITL while maintaining ≥ 0.99 recall on ABORT?

Faithful Explanation (C3): Three Levels of Verifiability

The Core Distinction

Plausible explanation: “I recommended spraying because conditions were suitable.”

Faithful explanation: “I recommended spraying because: (1) soil moisture = 41% < 45% threshold [AGROVOC:soil-moisture-threshold]; (2) wind = 1.8 m/s < 3 m/s spray limit [EPPO:spray-conditions]; (3) last spray was 6 days ago [audit-log:event-2847]; HITL triggered: action=SPRAY.”

Level 1: Rule-Auditable

HITL trigger reasons are explicit rule-derived strings. No LLM involved.

Level 2: KB-Traceable

Every command element cites PPDB/EPPO/AGROVOC source URI as verifiable provenance.

Level 3:

Attribution-Grounded

Integrated Gradients attribute which input tokens drove the LLM's intent prediction.

Connection to My BVF Research

BVF uses Fisher-Rao geometry to trace representation transitions across Llama/OLMo/Gemma/Mistral/Qwen. C3 applies the same analytical lens to the agricultural fine-tuned LLM: which contextual features (sensor readings, KB entities) activate each intent class in representation space. **BVF → C3 is a direct research lineage, not a stretch.**

Metaflow: Scientific Necessity, Not Engineering Preference

Why Reproducibility Is a Scientific Problem Here

- Agricultural AI experiments are weather-dependent, season-bound, farm-specific
- Published results cannot be replicated without complete pipeline versioning
- Supervisors in **Lyon and Milan** must inspect *identical* pipeline state

Metaflow's "3 Vs" (Tagliabue et al., MLSys 2023):

- **Velocity**: Iterate on C1 without re-running IoT ingestion
- **Validation**: Both supervisors inspect same artifact-level run
- **Versioning**: Model regression → detectable by run-ID comparison

Prototype → Production: Same Code

@batch(gpu=1) on fine-tuning step:

- Local: falls back to CPU
- AWS: uses GPU endpoint

**Pipeline validated in Lyon (M6–8)
deploys at UniMI farm in Milan (M20)
with zero code changes.**

DAG Concepts in This PhD

Business Logic = HITL rules (version-controlled)

Environment = @batch / @kubernetes / local

Documentation = @card → agronomist reports

Evaluation Plan: 9-Metric Framework

Dimension	Metric	Target
Intent classification	Macro-F1 on held-out agricultural corpus	≥ 0.92
Calibration	Expected Calibration Error (ECE)	< 0.05
Safety recall	Recall on ABORT class	≥ 0.99
Grounding fidelity	% commands with all entities KB-verified	≥ 0.95
Faithfulness	Attribution $\leftrightarrow$ KB trace alignment (Kendall τ)	> 0.70
Edge latency	Intent classification on Jetson Orin	< 800 ms
HITL fatigue	Confirmation rate on unambiguous routine queries	$< 15\%$
Trust calibration	Operator acceptance of correct recommendations	≥ 0.80
Spray accuracy (Y2)	Dosage error vs. prescription map	$< 5\%$

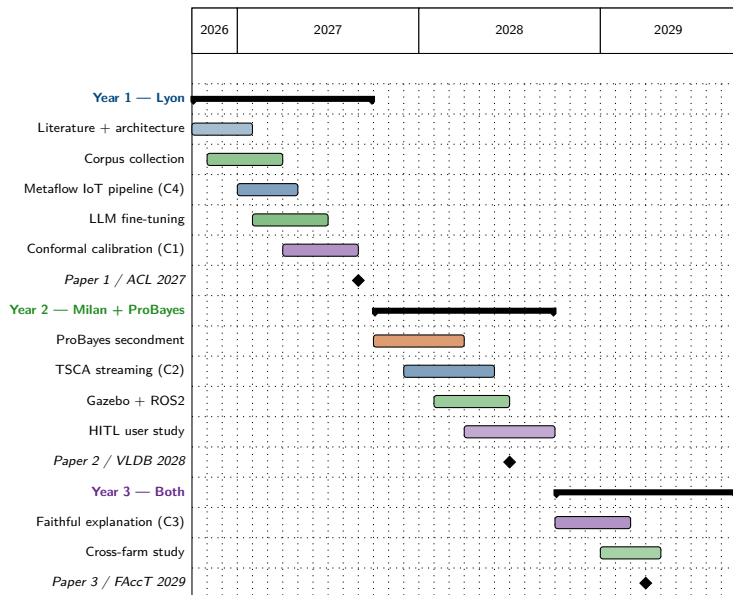
Datasets

- Agronomist interviews, ProBayes partner farms
- GPT-4 synthetic + expert validation (500+ queries)
- ACRE drone imagery, PANGAEA soil datasets
- OpenWeather API for meteorological context

Year 2 Simulation Baseline

All metrics except spray accuracy achievable in Gazebo simulation. Real robot validation (Prof. Oberti) in M14–20. **All 3 papers feasible on simulation + public data alone.**

36-Month Research Roadmap



Why I Am the Right Candidate

What I Am:

- Researcher in **trustworthy conversational AI** for human-supervised robotic systems
- Not: robotics hardware engineer
- Not: generic agriculture chatbot developer

BVF Research → C3 (Direct Lineage)

Fisher-Rao geometry traces LLM belief transitions in Llama, OLMo, Gemma, Mistral, Qwen. Same analytical lens → attribution-grounded explanation for agricultural intent classifier. **C3 is BVF applied to a high-stakes domain.**

CFL Paper → C2 Year 3

SpringerNature ICOMP'24: clustered federated learning for heterogeneous edge data. → Privacy-preserving cross-farm model generalisation

Teaching Experience → User Study Design

3 years: Nalhati Govt. Polytechnic, 50+ IoT student projects. → Practical IoT deployment realities. → Designing HITL user studies for non-expert operators (Year 2).

Supervision Fit

Dr. Vargas-Solar (Lyon): DAG data management + query formalisation for TSCA (C2). C1, C3 NLP/AI research base.

Prof. Oberti (Milan): Precision crop protection domain knowledge. Real robot integration Year 2.

Honest edge: Limited robotics hardware experience.

Plan: Year 1 simulation-first (Gazebo). Year 2 guided integration under Prof. Oberti.

This is a *learning trajectory*, not a gap.

Expected Contributions and Open Questions for You

Four Contributions × Four Venues:

- **C1 – CIC:** Conformal prediction for agricultural intent
ACL 2027 / EMNLP 2027
- **C2 – TSCA:** Streaming temporal grounding
VLDB 2027 / AAAI 2028
- **C3 – FAES:** Faithful attribution explanations
FAccT 2028
- **C4 – RAAI:** Metaflow for reproducible agri-AI
MLSys 2027 / VLDB Demo

One-sentence summary:

“How can calibrated uncertainty estimation and faithful explanation make NL-commanded agricultural robots trustworthy for non-expert human oversight?”

Open Questions for Supervisors

- ① **Dr. Vargas-Solar:** How does your DAG-based dataflow formalism map to the TSCA streaming consistency model? Are there query operators from your prior work that directly apply?
- ② **Prof. Oberti:** For the HITL user study in Year 2 — what operator roles and farm scenarios from your precision spraying work would be most informative for ecological validity?
- ③ **Both:** How do you envision the ProBayes secondment shaping the calibration methods in C1?

Thank you. I welcome your critical feedback on the proposal.